

FourierGNN: Rethinking Multivariate Time Series Forecasting from a Pure Graph Perspective

Omkar Bhandare (22CS30016) | Shivam Choudhury (22CS10072) | Ayush Rawat (22CS10012)

1 Problem Statement

Multivariate time series (MTS) forecasting is a fundamental problem in domains such as traffic prediction, weather modelling, and energy demand forecasting, where accurate predictions require capturing both **temporal dynamics** within individual variables and **spatial correlations** across variables. Existing deep learning approaches typically decompose this problem into two components: a temporal module to model intra-series dependencies, and a graph neural network to capture inter-series relationships. Though effective, the decomposition approach introduces architectural complexity, increases computational cost, and enforces an artificial separation between the spatial and temporal modelling. This separation often fails to represent the **spatiotemporal dependencies**, which are tightly coupled and often time-varying. The core problem addressed by the authors is therefore how to design a single, unified graph-based framework that can jointly model spatial and temporal dependencies in MTS while maintaining computational efficiency and scalability to large numbers of variables and time steps.

2 Previous Solutions and their Limitations

Previous Solutions

- **RNN/LSTM-based models:** Model temporal dependencies within each variable using sequential recurrence, achieving strong short-term forecasting but treating each series largely independently.
- **TCN-based models:** Use dilated convolutions to capture temporal patterns in parallel, improving efficiency but focusing primarily on intra-series temporal structure.
- **Transformer-based models (Informer, Autoformer, etc.):** Apply attention mechanisms to model long-range temporal dependencies, often with sparsity or frequency-domain approximations to reduce computational cost.
- **Spatiotemporal GNN hybrids (STGCN, DCRNN, GraphWaveNet, MTGNN, AGCRN, StemGNN):** Combine temporal encoders with GNNs to explicitly model spatial correlations across variables.
- **Fourier/frequency-domain temporal methods (SFM, FEDformer):** Leverage spectral representations to capture long-term temporal dependencies more efficiently than time-domain operations.

Limitations

- **Decoupled spatial-temporal Modelling:** Most methods treat temporal and spatial dependencies using separate modules, limiting their ability to capture unified spatiotemporal interactions.
- **High Computational Complexity:** Dense spatiotemporal modelling often incurs quadratic cost with respect to the number of variables and time steps.
- **Static/Predefined Graph Structures:** Many GNN-based methods rely on fixed adjacency matrices that fail to capture time-varying relationships.
- **Architectural Complexity:** Two-stage designs increase parameter count, training difficulty, and deployment overhead.

3 Novel Solution

The authors propose **FourierGNN**, a *pure-graph* framework for multivariate time series (MTS) forecasting that eliminates the conventional separation between spatial and temporal modelling. The key idea is to (i) represent each scalar observation as a node in a unified spatiotemporal graph and (ii) perform expressive graph convolutions efficiently in the Fourier domain.

Hypervariate graph construction

Given an input window $X_t \in \mathbb{R}^{N \times T}$ with N variables and window length T , FourierGNN constructs a *hypervariate graph* by treating every scalar value $x_{i,\tau}$ as an individual node. The resulting graph contains

$$n = N \cdot T \quad \text{nodes}, \quad X_t^G \in \mathbb{R}^{n \times 1}.$$

The adjacency matrix A_t^G is initialized as fully connected and learned end-to-end. This single graph uniformly captures (i) spatial dependencies (across variables), (ii) temporal dependencies (across time), and (iii) spatiotemporal dependencies (across both), without requiring separate spatial and temporal networks.

Fourier Graph Operator (FGO)

Direct graph convolution on the large hypervariate graph is computationally prohibitive. To address this, the authors define the *Fourier Graph Operator (FGO)* using a translation-invariant kernel κ . Let $\mathcal{F}(\cdot)$ denote the Discrete Fourier Transform (DFT); the operator is defined as

$$S_{A,W} := \mathcal{F}(\kappa).$$

By the convolution theorem, graph convolution in the time domain is equivalent to element-wise multiplication in the Fourier domain:

$$\mathcal{F}(X) S_{A,W} = \mathcal{F}(AXW). \quad (1)$$

For the fully connected hypervariate graph, the FGO is shared across nodes, yielding a parameter-efficient yet expressive operator.

Architecture and interpretation

FourierGNN stacks K FGOs directly in the Fourier domain:

$$\text{FourierGNN}(X, A) = \sum_{k=0}^K \sigma(\mathcal{F}(X) S_{0:k} + b_k), \quad S_{0:k} = \prod_{i=0}^k S_i.$$

The paper shows that recursively multiplying FGOs in Fourier space corresponds, after inverse DFT, to *multi-order graph convolutions* in the time domain:

$$\mathcal{F}^{-1}(\mathcal{F}(X) S_{0:k}) = A_{k:0} X W_{0:k}.$$

Thus, stacking FGOs explicitly models higher-order diffusion over the hypervariate graph.

Computational efficiency

A direct time-domain graph shift incurs $\mathcal{O}(n^2d + nd^2)$ complexity. In contrast, FourierGNN operates in

$$\mathcal{O}(nd \log n + nd^2),$$

reducing the dominant quadratic dependence on n to near log-linear. This enables scalable learning on large $N \times T$ spatiotemporal graphs.

4 How FourierGNN Addresses Previous Limitations

FourierGNN overcomes the main limitations of existing multivariate time series forecasting methods through a unified graph representation and efficient spectral computation.

Unified spatio-temporal modelling: By representing each variable–time pair as a node in a hypervariate graph, FourierGNN jointly models spatial and temporal dependencies within a single graph structure, eliminating the need for separate temporal encoders or staged spatio-temporal pipelines.

Learned and adaptive relations: The hypervariate graph is initialized as fully connected and learned end-to-end, enabling FourierGNN to capture task-specific and time-varying inter-variable relationships without relying on predefined or static graph structures.

Computational scalability: FourierGNN replaces expensive time-domain graph convolution with Fourier Graph Operators, reducing the dominant complexity from quadratic to near log-linear in the number of spatio-temporal nodes and enabling efficient learning on large MTS inputs.

Expressive multi-order interactions: Stacking Fourier Graph Operators implicitly realises higher-order graph diffusion, allowing the model to capture long-range and multi-step spatio-temporal dependencies while maintaining a compact parameterization.

5 Datasets

The proposed solution, FourierGNN, uses seven public multivariate benchmarks for multivariate time series forecasting and these benchmark datasets are summarised below

Dataset	Solar	Wiki	Traffic	ECG	Electricity	COVID-19	METR-LA
Samples	3650	803	10560	5000	140211	335	34272
Variables	592	2000	963	140	370	55	207
Granularity	1 hour	1 day	1 hour	- -	15 min	1 day	5 min
Train/Val/Test	70/20/10	70/20/10	70/20/10	70/20/10	70/20/10	60/20/20	70/20/10

Table 1: Summary of datasets

6 Results and Conclusion

Models	Solar			Wiki			Traffic		
	MAE	RMSE	MAPE(%)	MAE	RMSE	MAPE(%)	MAE	RMSE	MAPE(%)
VAR	0.184	0.234	577.10	0.057	0.094	96.58	0.535	1.133	550.12
SFM	0.161	0.283	362.89	0.081	0.156	104.47	0.029	0.044	59.33
LSTNet	0.148	0.200	132.95	0.054	0.090	118.24	0.026	0.057	25.77
TCN	0.176	0.222	142.23	0.094	0.142	99.66	0.052	0.067	- -
DeepGLO	0.178	0.400	346.78	0.110	0.113	119.60	0.025	0.037	33.32
Reformer	0.234	0.292	128.58	0.048	0.085	73.61	0.029	0.042	112.58
Informer	0.151	0.199	128.45	0.051	0.086	80.50	0.020	0.033	59.34
Autoformer	0.150	0.193	103.79	0.069	0.103	121.90	0.029	0.043	100.02
FEDformer	0.139	0.182	100.92	0.068	0.098	123.10	0.025	0.038	85.12
GraphWaveNet	0.183	0.238	603.00	0.061	0.105	136.12	0.013	0.034	33.78
StemGNN	0.176	0.222	128.39	0.190	0.255	117.92	0.080	0.135	64.51
MTGNN	0.151	0.207	507.91	0.101	0.140	122.96	0.013	0.030	29.53
AGCRN	0.123	0.214	353.03	0.044	0.079	78.52	0.084	0.166	31.73
FourierGNN	0.120	0.162	116.48	0.041	0.076	64.50	0.011	0.023	28.71

Models	ECG			Electricity			COVID-19		
	MAE	RMSE	MAPE(%)	MAE	RMSE	MAPE(%)	MAE	RMSE	MAPE(%)
VAR	0.120	0.170	22.56	0.101	0.163	43.11	0.226	0.326	191.95
SFM	0.095	0.135	24.20	0.086	0.129	33.71	0.205	0.308	76.08
LSTNet	0.079	0.115	18.68	0.075	0.138	29.95	0.248	0.305	89.04
TCN	0.078	0.107	17.59	0.057	0.083	26.64	0.317	0.354	151.78
DeepGLO	0.110	0.163	43.90	0.090	0.131	29.40	0.169	0.253	75.19
Reformer	0.062	0.090	13.58	0.078	0.129	33.37	0.152	0.209	132.78
Informer	0.056	0.085	11.99	0.070	0.119	32.66	0.200	0.259	155.55
Autoformer	0.055	0.081	11.37	0.056	0.083	25.94	0.159	0.211	136.24
FEDformer	0.055	0.080	11.16	0.055	0.081	25.84	0.160	0.219	134.45
GraphWaveNet	0.093	0.142	40.19	0.094	0.140	37.01	0.201	0.255	100.83
StemGNN	0.100	0.130	29.62	0.070	0.101	- -	0.421	0.508	141.01
MTGNN	0.090	0.139	35.04	0.077	0.113	29.77	0.394	0.488	88.13
AGCRN	0.055	0.080	11.75	0.074	0.116	26.08	0.254	0.309	83.37
FourierGNN	0.052	0.078	10.97	0.051	0.077	24.28	0.123	0.168	71.52

Table 2: Overall performance of forecasting models on the six datasets.

FourierGNN consistently attains the best or near-best MAE and RMSE across the evaluated datasets. Averaging across datasets, FourierGNN yields notable reductions in error (reported in the paper as 9.4% MAE and 10.9% RMSE improvement over the best baselines).

7 Future Works

7.1 Limitations of FourierGNN

- **Loss of Sequential Information:** Treating every time-step as a node makes the graph *order-insensitive*. Permuting the time sequence results in an identical graph structure, losing temporal context.
- **Attention Dilution:** The Fully-Connected structure treats all connections uniformly. This “over-smooths” the data, washing out distinct local patterns with global noise.

7.2 What we think?

Proposal: Replace Hypervariate Graphs with **Hyperspectral Graphs**.

- **The Method:** Embeds sequential order into frequency signals *before* graph construction.
- **Intuitive Benefit:**
 - *Preserves Time:* Embedding frequency locks the sequential order.
 - *Sharper Focus:* Allow the model to capture the global trends without blurring out specific, local features.

8 Contributions

Ayush Rawat - Report, Dataset Sourcing, Code

Omkar Bhandare - Code, Future Directions

Shivam Choudhury - Dataset Sourcing, Future Directions, Code